

Working Groups (all)

GY457 Autumn Term 2025 (Seminar Topic 6)

Lecturer: *Dr Sara Bagagli*

Context¹

This seminar builds on the lecture on Topic 6: “Firm location choice.” During the lecture, we have generalized the concept of bid-rents to commercial uses. We also discussed the role of agglomeration economies in facilitating more complex spatial structures. One of the phenomena we discussed are edge cities, which are satellite towns developed by profit-maximizing developers. The purpose of this seminar is to deepen the understanding of these concepts and to explore the trade-offs a developer faces in choosing a location of an edge city in a numerical example. The idea is to illustrate how urban economics theory and evidence help making good financial decisions on real estate markets.

“Homework” to be carried out prior to the seminar

The compulsory reading for this seminar is:

Henderson V., and Mitra, A. (1996): The new urban landscape: Developers and edge cities. *Regional Science and Urban Economics* 26, 613-643.

All students are expected to read this paper. To answer the seminar questions, it is recommended that students revisit the following paper as it contains useful information regarding the parameter choice in the numerical exercise.

Ahlfeldt, G., and Pietrostefani, E. (2019). The Economic Effects of Density: A Synthesis. *Journal of Urban Economics*, 111: 93-107.

All students must be prepared to answer the questions during the seminar. It is recommended that they execute the numerical exercise and prepare some notes. However, no submission of written answers is expected. *Presenters*, in addition, are expected to prepare slides for a joint presentation of their answers (one presentation per seminar group). In doing so, they may split the tasks or work together on all tasks as they prefer.

Seminar tasks:

- 1) In 1991, Garreau established the concept of edge cities. What are the characteristics of edge cities and what makes them attractive to firms and residents?

¹ The seminar was originally designed by Prof. Ahlfeldt and is kindly used with permission.

- 2) Henderson and Mitra (1996) develop a model of optimal edge city location choice. With reference to their model, engage with the following tasks:
 - a Discuss the concept of monopsony power and its relevance to the developer in the context of edge city development
 - b What trade-off does the developer face when making a location choice for the edge city? Which forces push towards a peripheral location or a more central location with respect to the core city?
 - c If the core city is larger, should the developer develop a larger or smaller edge city?
 - d Following the model predictions, should we think of the edge city location choice as “random” or “chaotic”?
- 3) Let’s now consider a simpler framework in which a developer seeks to make a profit-maximizing decision on where to construct the edge city. To address the tasks below, please open the GY457_Topic_6_Seminar_Data.xlsx document. As evident from the parameters given in the file, the developer is committed to develop an edge city on a 40,000 m² land parcel at a floor area ratio of 1.25. Their internal rate of return ([a brief description](#)) is 6% and they face a per-square meter office space construction cost of 2,500 (in an arbitrary currency). The developer considers 11 locations at different distances from the CBD. They observe the local land prices, at which they have to buy the land, as well as the current employment densities. They do not observe the office rents that can be realized at those locations. You, as consultants, have been commissioned to help the developer choosing the profit-maximizing location for the edge city development.
 - a. To predict the market rent at the considered locations, you have collected a data set with per-square meter office rents for a large number of buildings. Along with the rent, you have obtained information on the location of those buildings relative to the CBD. Use a scatter plot ([example](#)) to inspect the data and use [regression analysis](#) to predict the office rents at the considered location (there are many helpful introductions into simple regression analysis if you have not done it much before, see e.g. for how to do it in [Excel](#) or [Stata](#)). Please carefully inspect the scatter plot to find the appropriate functional form. If helpful to reduce noise, you may also try [binscatter](#) command in Stata. How does that compare to a simple scatterplot? The data set is in the worksheet “Observed rent data.”
 - b. Using the imputed office rents by location, calculate for every location in the “Working sheet” the yearly office rent per square meter of land, the net present value ([NPV](#)) of future rents per square meter of land, the construction cost per square meter of land, and the profit per square meter of land. Check the notes in the “Working sheet” for hints on how to compute the variables. At this stage, assume that the development of

the edge city has no effect on the office rent at the locations. Which location would you recommend to the developer?

- c. Now, assume that the added employment that comes with the edge city generates agglomeration economies that have a positive effect on rents. Assume that at each location (each one is a spatial unit with an area of one square kilometre) there is enough land available so that the added employment does not crowd out existing employment. For each location, compute the ratio of the employment density with the edge city over the employment density without the edge city. With the help of an appropriate elasticity (check the readings and lecture notes), compute the ratio of the rent achievable with the added agglomeration effect due to the edge city over the rent without the edge city effect. Compute the yearly office rent per square meter land, the NPV of future rents per square meter, the profit per land unit, and the total profit. Which location would you now recommend to the developer?
- 4) Competitive land markets imply that excess profits capitalize into land prices, i.e. the net present value of an investment in per land-unit terms determines the land price. Brueckner (1987) nicely summarizes this equilibrium relationship. The implication is that developers should not make large profits in excess of the typical internal rate of return. Use the results from the case study and your wide knowledge of urban economics and welfare economics to answer the following questions:
 - a. Why can the edge city developer make a large profit on the development in the edge city?
 - b. If there is so much “money on the table”, why doesn’t perfect competition for land imply that the expected profit is capitalized in the land price?
 - c. Why is the emergence of edge cities a relatively recent phenomenon?

During the seminar

After an introduction and discussion of initial questions, the presenters will walk the group through their answers to the questions in an “interactive” presentation. All presenters should carry an approximately similar weight in the presentation. During the presentation, it is expected that the presenters “engage” with the audience, i.e. prompt the audience to suggest answers. Likewise, the audience is expected to challenge the answers provided by the presenters.

The presentation, in total, may cover up to 75 minutes. To allow for sufficient time for group discussions and clarifications, the net-presentation time must be shorter. The lecturer will monitor the presentation and the discussion and intervene if appropriate. The presenters are expected to make notes during the discussion in case corrections to their suggested answers are due. After the presentation, the investors (the audience) will cast a vote to reveal whether or not they have been convinced by the investment

strategy. We may move on to a more general discussion about the pros and cons of the suggested (and related) investment strategies.

After the seminar

If indicated by the feedback during the seminar, presenters will revise their presentation. Within 24 hours, they will then ask the lecturer to upload it onto the GY457 Moodle page. Please, use the following filenames to save your presentations: Topic_6_group_X, where X indicates the seminar group.